

8.2 Classwork: Rational Functions — Toward the Maturità

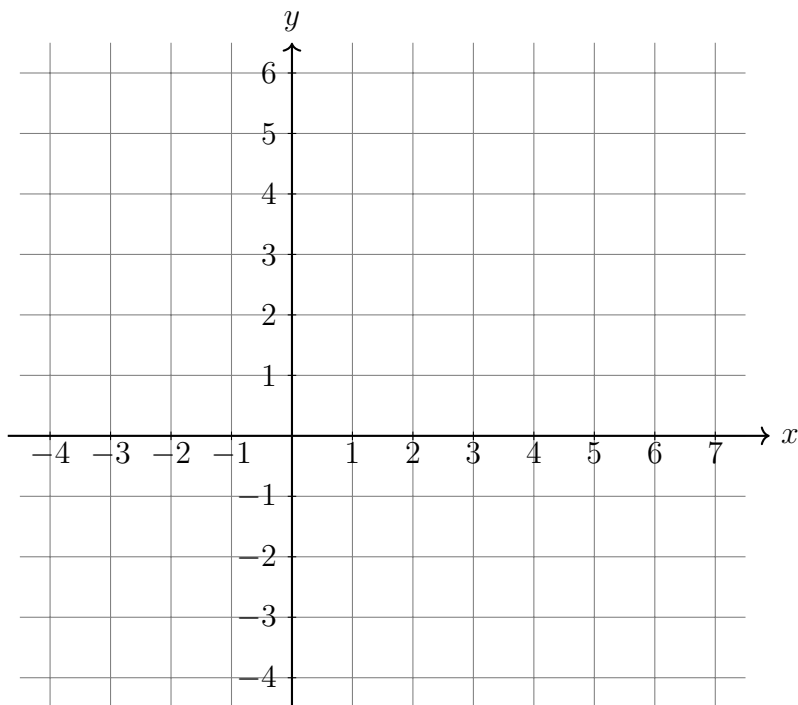
1. Consider $f(x) = \frac{x+3}{x-2}$.

(a) State the equation of the vertical asymptote.

(b) Verify by long division (or otherwise) that $f(x) = 1 + \frac{5}{x-2}$. Use this form to state the equation of the horizontal asymptote.

(c) Find the x -intercept and the y -intercept.

(d) Sketch f on the axes below. Mark and label the asymptotes and intercepts.



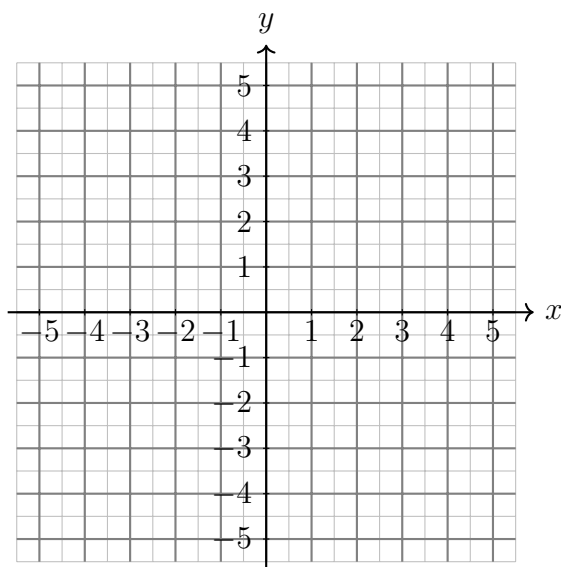
2. Consider the family of functions $f_a(x) = \frac{x-2}{x-a}$, with $a \neq 2$.

(a) For an arbitrary value of a , state the equation of the vertical asymptote of f_a .

(b) Find the horizontal asymptote of f_a . Does its equation depend on a ?

(c) Compute $f_a(2)$. What does this tell you about a point shared by every curve in the family?

(d) Now set $a = -1$. Find the y -intercept of f_{-1} and sketch its graph on the axes below. Mark the asymptotes (dashed) and the intercepts.



3. Consider $g(x) = \frac{x^2 + x}{x^2 + 1}$.

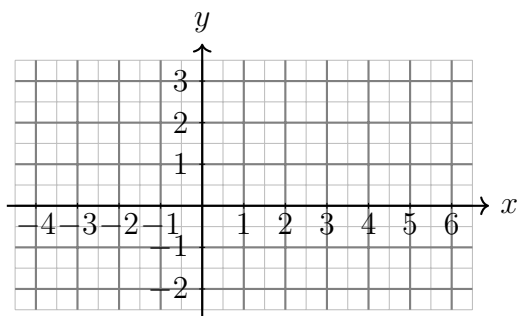
(a) State the domain of g . (Hint: look at the denominator carefully — can it equal zero?)

(b) Compute $\lim_{x \rightarrow +\infty} g(x)$ and $\lim_{x \rightarrow -\infty} g(x)$. Hence state the horizontal asymptote.

(c) Find the x -intercepts (note: there are two) and the y -intercept.

(d) Complete the table of values, then sketch the graph below.

x	-3	-2	-1	0	1	2	3	5
$g(x)$								



4. Consider the family of functions $f_a(x) = \frac{x^2 - ax}{x^2 - a}$, with $a \neq 0$.

The function g from Problem 3 is the member f_{-1} (check: substituting $a = -1$ gives $g(x) = (x^2 + x)/(x^2 + 1)$).

(a) Find the domain of f_a in each of the following cases.

(i) $a < 0$ (for example $a = -1$, $a = -3$):

(ii) $a > 0$ (for example $a = 4$):

(b) State the equation of the horizontal asymptote of f_a . (Hint: same idea as Problem 3(b) — compare leading terms.)

(c) Set $f_a(x)$ equal to the horizontal asymptote you found in (b), and solve for x . Show that the solution does not depend on a (provided $a \neq 1$). Conclude that every curve Ω_a in the family passes through one common point on its horizontal asymptote. State this point.

(d) Compute $f_a(0)$. Then show, using the quotient rule, that $f'_a(0) = 1$ for every $a \neq 0$. Hence write down the equation of the line tangent to Ω_a at the origin. Is this line the same for every a ?

5. Consider $h(x) = \frac{x^3 + 4}{x^2}$.

(a) State the domain of h , and the equation of the vertical asymptote.

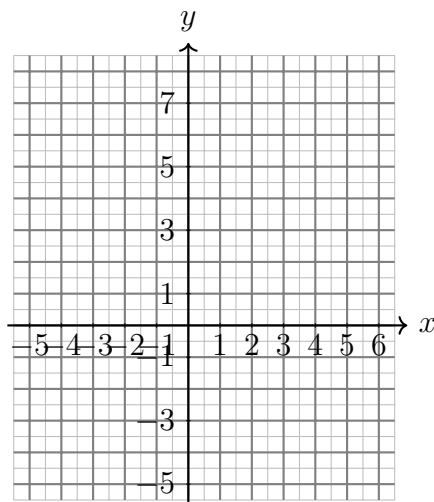
(b) Show that $h(x) = x + \frac{4}{x^2}$.

(c) Use part (b) to explain why the graph of h approaches the line $y = x$ as $|x|$ grows large. This line is called the *oblique asymptote* of h .

(d) Complete the table of values for h .

x	-3	-2	-1	-0.5	0.5	1	2	4
$h(x)$								

(e) Sketch the graph on the axes below. Show both the vertical asymptote and the oblique asymptote $y = x$ as dashed lines.



Problemi della Maturità

(Use separate paper for these problems.)

6. PROBLEMA 2 — Sessione ordinaria 2023.

Fissato un parametro reale a , con $a \neq 0$, si consideri la funzione f_a così definita:

$$f_a(x) = \frac{x^2 - ax}{x^2 - a}$$

il cui grafico sarà indicato con Ω_a .

- (a) Al variare del parametro a , determinare il dominio di f_a , studiarne le eventuali discontinuità e scrivere le equazioni di tutti i suoi asintoti.
- (b) Mostrare che, per $a \neq 1$, tutti i grafici Ω_a intersecano il proprio asintoto orizzontale in uno stesso punto e condividono la stessa retta tangente nell'origine.

7. PROBLEMA 1 — Sessione ordinaria 2024.

Si consideri $f_{a,b}(x) = \frac{ax^3 + b}{x^2}$, con $a, b \in \mathbb{R}$.

- (a) Determinare i valori dei parametri in modo che la retta t , di equazione $7x + y - 12 = 0$, sia tangente al grafico di $f_{a,b}(x)$ nel suo punto P di ascissa $x = 1$.

Si ponga, d'ora in avanti, $a = 1$ e $b = 4$.

- (b) Studiare la funzione $f(x) = \frac{x^3 + 4}{x^2}$ e tracciarne il grafico γ .